Equivalence covers of powers of cycles when $k+1$ is a power of two

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Abstract

Let C_n^k be the k-th power of the cycle C_n and let $\mathrm{eq}(G)$ be the least number of equivalence graphs (vertex-disjoint unions of cliques) whose union is G. Yang (arXiv:2610.15635) showed that for large n, $\mathrm{eq}(C_n^k) \in \{m+1, m+2\}$ when $k = 2^m - 1$. Yang also determined the value when $(k+1) \mid n$ (it is $m+1$) and when n is odd (it is $m+2$). The case of even n not divisible by $k+1$ was left open. We prove a structural lemma: in a cover with $m+1$ layers, every class has span exactly k, and certain weighted pairs are covered exactly once. Using it, we settle the open case completely for $k = 3$ and $k = 7$, for all $n \geq 2k+2$: $\mathrm{eq}(C_n^k) = m+1$ if $(k+1) \mid n$ and $m+2$ otherwise. The proof is computer-assisted: we compute a finite transfer graph and show that all of its cycles have length divisible by $k+1$, and we certify small covers with SAT. We conjecture that the same holds for every m.

1 Introduction

An *equivalence graph* is a vertex-disjoint union of cliques. The *equivalence covering number* $\mathrm{eq}(G)$ is the least t such that the edges of G are covered by t equivalence subgraphs of G (Alon [1]). We call these subgraphs *layers* and their cliques *classes*. We take each layer to be a partition of $V(G)$, allowing singleton classes.

For C_n^k with $n \geq 2k+2$, the vertices are $\mathbb{Z}_n$ and $u \sim v$ iff $1 \leq |u-v|_n \leq k$. Write $N = k+1$. Yang [2] proved:

- $1 + \lceil \log_2 N \rceil \leq \mathrm{eq}(C_n^k) \leq 1 + 2\lceil \log_2 N \rceil$ for $n \geq 2k+2$;
- $\mathrm{eq}(C_n^k) \in \{1 + \lceil \log_2 N \rceil,\ 2 + \lceil \log_2 N \rceil\}$ for $n \geq N \max\{41, 5\lceil \log_2 N \rceil\}$;
- for $n \geq \max\{2k+2, k^2(k+2)\}$, $\mathrm{eq}(C_n^k) = 1 + \lceil \log_2 N \rceil$ if N is not a power of two or $N \mid n$, and $= 2 + \lceil \log_2 N \rceil$ if N is a power of two and n is odd.

So when $N = 2^m$ and n is even with $N \nmid n$, the value is $m+1$ or $m+2$, but which one was not decided. This note addresses that case. Our main results:

Theorem 1. *Let $k \in \{3, 7\}$, $N = k+1 = 2^m$ and $n \geq 2k+2$. Then $\mathrm{eq}(C_n^k) = m+1$ if $N \mid n$, and $\mathrm{eq}(C_n^k) = m+2$ otherwise.*

Conjecture 2. *The statement of Theorem 1 holds for every $k = 2^m - 1$ with $m \geq 2$.*

The proof of Theorem 1 combines a structural lemma valid for all m (Section 2) with a finite computation for $m = 2, 3$ (Section 3), SAT-certified covers for small n, and Yang's results for large n (Section 4).

*Proposed the research direction and asked for a problem that was still open.

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2 A weighting and the structure of $(m+1)$-covers

Fix $m \geq 2$, $N = 2^m$, $k = N - 1$. Define weights on distances $d \in \{1, \ldots, k\}$:

$$w(N - 2^j) = 1 \ (1 \leq j \leq m-1), \qquad w(N-1) = 2, \qquad w(d) = 0 \text{ otherwise.}$$

For a finite set S of integers, put $w(S) = \sum_{\{x,y\} \subseteq S} w(|x - y|)$.

Lemma 3 (Weight lemma)**.** *Let $S \subseteq \{a, a+1, \ldots, a+k\}$ be nonempty. Then $w(S) \leq |S|$. If equality holds, then $a \in S$ and $a + k \in S$.*

Proof. Shift so that $a = 0$. Every distance with positive weight has the form $N - 2^j$ with $0 \leq j \leq m-1$. If $x < y$ in S and $y - x = N - 2^j$, then $x \leq 2^j - 1 \leq N/2 - 1$ and $y \geq N - 2^j \geq N/2$. So weighted pairs join the lower part $P = S \cap [0, N/2)$ to the upper part $Q = S \cap [N/2, N)$. Encode $x \in P$ by $\alpha(x) = x$ and $y \in Q$ by $\beta(y) = N-1-y$; both take values in $[0, N/2)$. Then $y - x = N - 2^j$ iff $\alpha(x) + \beta(y) + 1 = 2^j$.

Let H be the graph on $\{0, 1, \ldots, N/2 - 1\}$ in which $a \neq b$ are adjacent iff $a + b + 1$ is a power of two. Every $a \geq 1$ has exactly one neighbour $b < a$: we need $2^j \in [a+1, 2a]$, and there is exactly one such power of two. Hence H is a tree rooted at 0. Now let $\widetilde{H}$ be the bipartite graph with two copies $\{a'\}$ and $\{a''\}$ of the vertex set, where $a'b''$ is an edge iff $a + b + 1$ is a power of two (this includes $a = b = 0$). Without the edge $0'0''$, $\widetilde{H}$ is two disjoint copies of the tree H, because H is bipartite. The edge $0'0''$ joins the two copies, so $\widetilde{H}$ is a tree.

The pairs of S at distance $N - 2^j$ correspond exactly to the edges of $\widetilde{H}$ inside the vertex set $X = \{\alpha(x)' : x \in P\} \cup \{\beta(y)'' : y \in Q\}$, and $|X| = |S|$. The pair $\{0, N-1\}$ corresponds to the edge $0'0''$ and has weight 2; every other weighted pair has weight 1. An induced subgraph of a tree is a forest, so it has at most $|X| - 1$ edges. Therefore

$$w(S) \leq (|S| - 1) + [\, 0 \in S \text{ and } N - 1 \in S \,] \leq |S|.$$

If $w(S) = |S|$, the bracket equals 1, so $0, N - 1 \in S$. □

Call S *tight* if $w(S) = |S|$. A computer count gives 4, 36 and 1764 tight sets inside $[0, k]$ for $m = 2, 3, 4$.

Corollary 4. *Let $n \geq 2k + 2$. Then $\mathrm{eq}(C_n^k) \geq m + 1$. In any cover of C_n^k by $m + 1$ layers:*

1. *every class is tight, so it has the form $v + S$ with $S \subseteq [0, k]$ tight, $0, k \in S$;*
2. *there are no singleton classes, and each vertex lies in exactly one class of each layer;*
3. *every pair at a distance d with $w(d) > 0$ is covered exactly once;*
4. *for every $v \in \mathbb{Z}_n$ there is exactly one class whose left end is v.*

Proof. Since $n \geq 2k+2$, the cyclic distance of two vertices within distance k is unambiguous. Each class lies in an arc of $k + 1$ consecutive vertices. The total weight to be covered is $n \sum_d w(d) = n(m+1)$. A single layer covers weight at most $\sum_{\text{classes}} |S| = n$ by Lemma 3. Hence at least $m + 1$ layers are needed. With exactly $m + 1$ layers, all these inequalities are equalities: every class is tight, and every weighted pair is covered exactly once. A singleton has weight $0 < 1$, so it is not tight. Every pair $\{v, v + k\}$ has positive weight and is covered exactly once, and every class is of the form $v + S$ with $0, k \in S$. So the class containing $\{v, v + k\}$ is the unique class with left end v. □

The lower bound $m + 1$ is already in [2]. The new content is the rigid structure in items 1–4.

3 The transfer graph

By Corollary 4, an $(m+1)$-layer cover of C_n^k determines a cyclic sequence $(\tau_v)_{v\in\mathbb{Z}_n}$ of *types* $\tau_v = (S_v, \ell_v)$, where $v + S_v$ is the class with left end v and $\ell_v \in \{0, \dots, m\}$ is its layer. Consider a vertex v. Every class containing v, and every pair $\{v-d, v\}$ with $d \le k$, involves only the types $\tau_{v-k}, \dots, \tau_v$. So the cover is valid at v iff the window $(\tau_{v-k}, \dots, \tau_v)$ satisfies these *local conditions*:

(a) for each layer ℓ, exactly one class $u + S_u$ with $\ell_u = \ell$ and $u \in [v-k, v]$ contains v;

(b) for each $1 \le d \le k$, at least one class contains $\{v-d, v\}$, and exactly one when $w(d) > 0$.

Let G_m be the directed graph whose vertices are k-tuples of types, with an arc $(\tau_0, \dots, \tau_{k-1}) \to (\tau_1, \dots, \tau_k)$ whenever $(\tau_0, \dots, \tau_k)$ satisfies (a) and (b). An $(m+1)$-layer cover of C_n^k gives a closed walk of length n in G_m. A closed walk stays inside one strongly connected component, and its length is a multiple of the component's period (the gcd of its cycle lengths). Hence:

Lemma 5. *If every strongly connected component of G_m that contains a cycle has period divisible by N, then C_n^k has no $(m+1)$-layer cover unless $N \mid n$.*

Computation. A k-tuple of types that lies on a cycle of G_m occurs, for every P, as the middle part of a sequence of P types satisfying (a) and (b) at every vertex whose window lies inside the sequence (unroll the cycle). For fixed P and offset c, we listed all k-tuples that occur at positions $c, \dots, c+k-1$ of such a sequence. This used a SAT solver (CaDiCaL via PySAT), cardinality constraints, and blocking clauses to enumerate every solution. Restricted to these tuples, G_m still contains every cycle of G_m. We built the arcs between them directly from (a) and (b), removed vertices with no incoming or outgoing arcs, and computed the strongly connected components and their periods.

- $m = 2$ ($k = 3$; 12 types; $P = 24$, $c = 8$): 12 tuples, forming 3 components. Each component is a single directed cycle of length 4. A direct enumeration of all 150 windows satisfying (a) and (b) gives the same graph. Up to permuting the three layers, the pattern alternates the tight sets $\{0, 3\}$ and $\{0, 1, 2, 3\}$, with layers rotating.

- $m = 3$ ($k = 7$; 144 types; $P = 48$, $c = 20$, and again with $P = 64$, $c = 28$): 288 tuples, forming 36 components. Each component is a single directed cycle of length 8. Ignoring layers, the tight sets follow one of three patterns of period 4 or 2. One is $[0, 7], \{0, 7\}, \{0, 1, 6, 7\}, \{0, 7\}$. The other two alternate $\{0, 7\}$ with $\{0, 1, 4, 5, 6, 7\}$, or with its mirror image $\{0, 1, 2, 3, 6, 7\}$. The layers then force period 8.

In both cases every cyclic component has period N. So Lemma 5 applies. Conversely, a closed walk satisfies (a) and (b) at every vertex and is therefore an $(m+1)$-layer cover. Each layer is a partition into cliques by (a), and every edge is covered by (b). So each cycle of length N, repeated n/N times, gives an $(m+1)$-layer cover of C_n^k for every $n \ge 2k+2$ with $N \mid n$.

4 Proof of Theorem 1

Lower bound. By Corollary 4, $\mathrm{eq} \ge m+1$. If $N \nmid n$, Lemma 5 and the computation rule out $m+1$ layers, so $\mathrm{eq} \ge m+2$.

Upper bound. If $N \mid n$, the periodic covers from Section 3 give $\mathrm{eq} \le m+1$ for all $n \ge 2k+2$. If $N \nmid n$, Yang's two-value theorem gives $\mathrm{eq} \le m+2$ for $n \ge N \max\{41, 5m\}$, that is, $n \ge 164$

for $k = 3$ and $n \geq 328$ for $k = 7$. For the remaining n ($8 \leq n \leq 163$ for $k = 3$, $16 \leq n \leq 327$ for $k = 7$), a SAT solver found explicit covers with $m + 2$ layers. In fact we found covers with the exact number of layers for every n in $[8, 163]$ and $[16, 440]$ respectively. Each cover was checked by an independent program: every layer is a disjoint union of cliques, and every edge is covered. The certificates are in the repository. □

Remark 6. For $m = 2, 3$ only a few local patterns survive, and in each of them the layer labels repeat only after N steps, even when the sets repeat sooner. A conceptual explanation of this for all m would prove Conjecture 2. For $m = 4$ there are 1764 tight sets and 5 layers; our SAT enumeration found no solution within two hours, so this case needs a better method.

Code and data

All scripts, logs and the cover certificates are in the accompanying repository: `tight.py` (tight sets), `satscan.py` (transfer graph; `layerscan.py` is a direct check for $k = 3$), `eqsat.py` and `upper.py` (SAT covers and their checks), `witness/` (certificates).

Use of AI

This work was carried out by an AI model (Claude, by Anthropic) at the request of Uchral Nyam-Osor, who proposed looking for a problem that nobody had yet solved. Uchral Nyam-Osor did not do the mathematics. The text has not been checked by a human expert. Corrections are welcome.

References

[1] N. Alon, Covering graphs by the minimum number of equivalence relations, *Combinatorica* 6 (1986) 201–206.

[2] Z. Yang, Covering powers of cycles with equivalence subgraphs, arXiv:2610.15635 [math.CO] (October 2026).